

Class Activity
Date:20-07-2026

1.Hospital Patient Admission Analysis

Code:

```
department <- c("Cardiology","Neurology","Pediatrics","Orthopedics","Emergency")
admissions <- c(120,90,150,110,180)
data <- data.frame(department, admissions)
barplot(data$admissions,
        names.arg=data$department,
        col="skyblue",
        main="Hospital Patient Admissions",
        xlab="Department",
        ylab="Number of Admissions")
admit <- matrix(c(60,40,80,50,90,
                  60,50,70,60,90),
               nrow=2, byrow=TRUE)
barplot(admit,
        beside=FALSE,
        col=c("blue","green"),
        names.arg=department,
        main="Stacked Bar Chart of Admissions",
        xlab="Department",
        ylab="Admissions")
legend("topright",
        legend=c("Male","Female"),
        fill=c("blue","green"))
install.packages("treemap")    # Run only once
library(treemap)
treemap(data,
        index="department",
        vSize="admissions",
        title="Hospital Department Distribution")
boxplot(admissions,
        main="Admission Variability",
        ylab="Admissions",
        col="orange")
hist(admissions,
        col="pink",
```

```

main="Distribution of Admissions",
xlab="Admissions",
ylab="Frequency")

```

Output:



2.E-Commerce Customer Spending Behaviour

Code:

```

Amount <- c(200,450,300,600,800,250,500,700,350,900)
Order_Frequency <- c(2,5,3,6,8,2,5,7,4,9)
Category <- c("Electronics","Fashion","Fashion","Electronics","Grocery",
              "Fashion","Electronics","Grocery","Fashion","Electronics")
data <- data.frame(Amount, Order_Frequency, Category)
hist(data$Amount,
     col="skyblue",
     main="Customer Spending Distribution",
     xlab="Purchase Amount",
     ylab="Frequency")
install.packages("vioplot") # Run only once
library(vioplot)
vioplot(Amount ~ Category,
       data=data,
       col="lightgreen",
       main="Spending by Category",
       xlab="Category",

```

```

      ylab="Purchase Amount")
boxplot(Amount ~ Category,
      data=data,
      col=c("pink","orange","lightblue"),
      main="Spending Variability",
      xlab="Category",
      ylab="Purchase Amount")
plot(density(data$Amount),
      col="blue",
      lwd=2,
      main="Density Plot of Customer Spending",
      xlab="Purchase Amount")
avg <- aggregate(Amount ~ Category, data, mean)
barplot(avg$Amount,
      names.arg=avg$Category,
      col=c("red","green","blue"),
      main="Average Spending by Category",
      xlab="Category",
      ylab="Average Purchase Amount")

```

Output:



3.University Examination Performance Dashboard

Code:

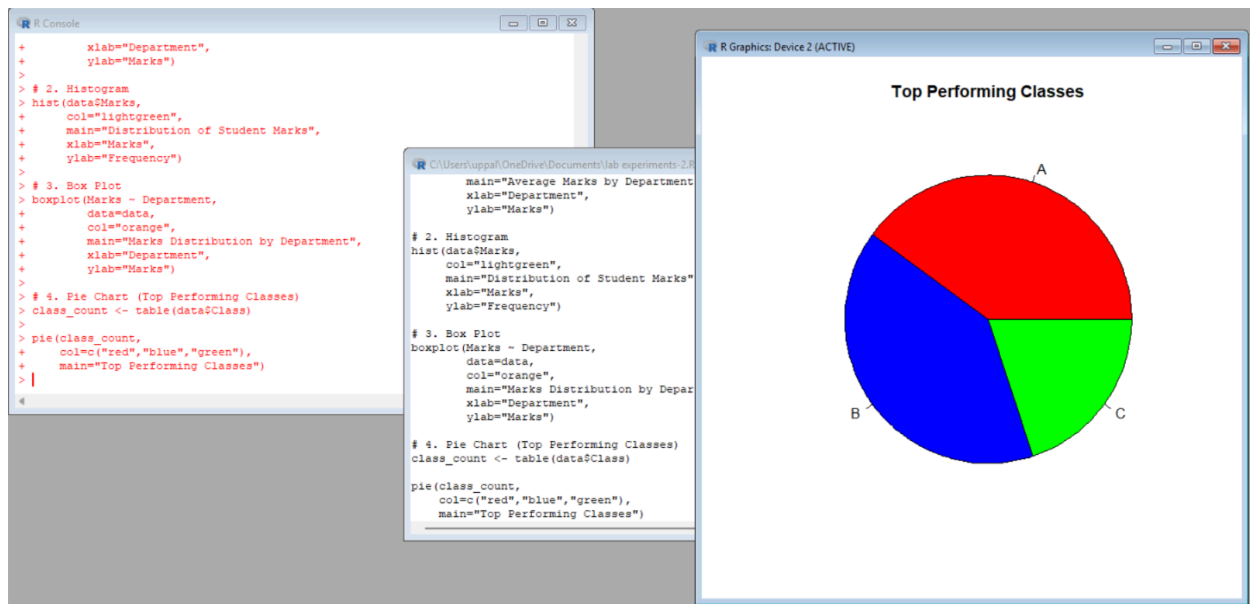
```
Department <- c("CSE","ECE","MECH","EEE","CIVIL")
```

```

Marks <- c(85,72,90,68,80)
Class <- c("A","B","A","C","B")
data <- data.frame(Department, Marks, Class)
barplot(data$Marks,
        names.arg=data$Department,
        col="skyblue",
        main="Average Marks by Department",
        xlab="Department",
        ylab="Marks")
hist(data$Marks,
     col="lightgreen",
     main="Distribution of Student Marks",
     xlab="Marks",
     ylab="Frequency")
boxplot(Marks ~ Department,
        data=data,
        col="orange",
        main="Marks Distribution by Department",
        xlab="Department",
        ylab="Marks")
class_count <- table(data$Class)
pie(class_count,
     col=c("red","blue","green"),
     main="Top Performing Classes")

```

Output:



4.Climate and Rainfall Distribution Study

Code:

```
City <- c("Chennai","Delhi","Mumbai","Bangalore","Hyderabad")
```

```
Rainfall <- c(120,80,250,150,100)
```

```
Temperature <- c(34,40,30,28,33)
```

```
Humidity <- c(75,45,85,70,60)
```

```
data <- data.frame(City, Rainfall, Temperature, Humidity)
```

```
hist(data$Rainfall,
```

```
  col="skyblue",
```

```
  main="Rainfall Distribution",
```

```
  xlab="Rainfall (mm)",
```

```
  ylab="Frequency")
```

```
plot(density(data$Rainfall),
```

```
  col="blue",
```

```
  lwd=2,
```

```
  main="Density Plot of Rainfall",
```

```
  xlab="Rainfall (mm)")
```

```
boxplot(Rainfall ~ City,
```

```
  data=data,
```

```
  col="lightgreen",
```

```
  main="Rainfall Variation by City",
```

```
  xlab="City",
```

```
  ylab="Rainfall (mm)")
```

```
barplot(data$Rainfall,
```

```
  names.arg=data$City,
```

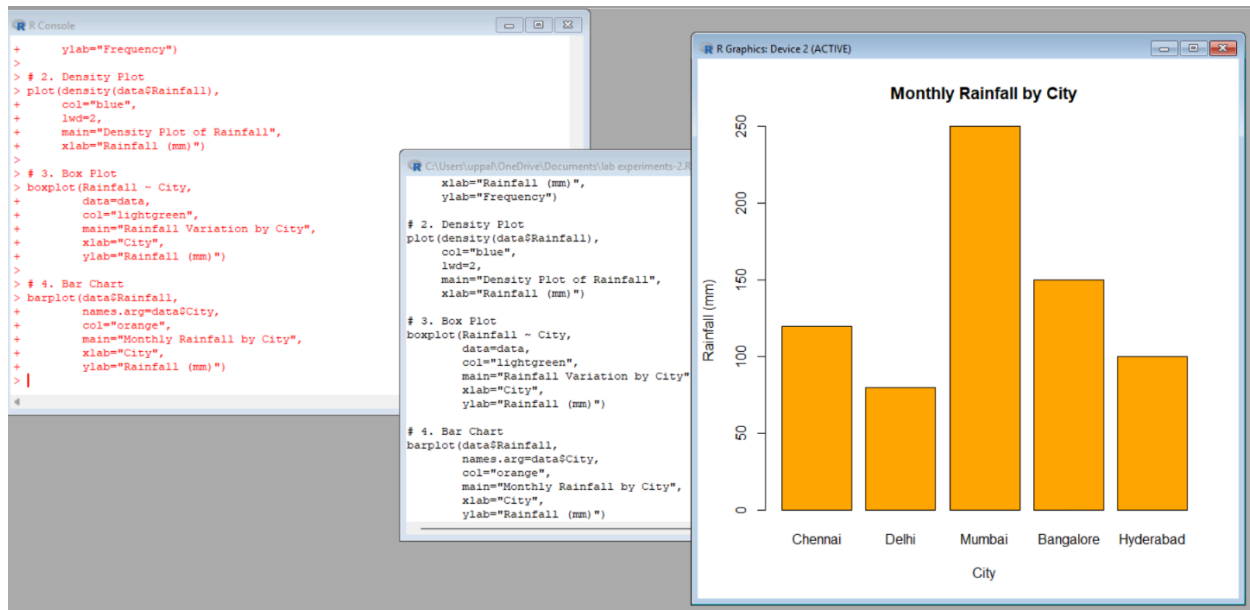
```
  col="orange",
```

```
  main="Monthly Rainfall by City",
```

```
  xlab="City",
```

```
  ylab="Rainfall (mm)")
```

Output:



5. Bank Loan Risk Assessment

Code:

```

Loan_Amount <- c(50000,100000,75000,120000,90000)
Income <- c(30000,60000,45000,80000,55000)
Credit_Score <- c(650,780,700,820,720)
Repayment <- c("Good","Excellent","Average","Excellent","Good")
data <- data.frame(Loan_Amount, Income, Credit_Score, Repayment)
hist(data$Loan_Amount,
     col="skyblue",
     main="Loan Amount Distribution",
     xlab="Loan Amount",
     ylab="Frequency")
boxplot(data$Loan_Amount,
     col="orange",
     main="Loan Amount Outliers",
     ylab="Loan Amount")
plot(density(data$Loan_Amount),
     col="blue",
     lwd=2,
     main="Density Plot of Loan Amount",
     xlab="Loan Amount")
replay <- table(data$Repayment)
barplot(replay,
     col=c("green","yellow","red"),

```

```

    main="Customer Repayment History",
    xlab="Repayment Status",
    ylab="Number of Customers")
plot(data$Income,
     data$Credit_Score,
     pch=19,
     col="red",
     main="Income vs Credit Score",
     xlab="Income",
     ylab="Credit Score")

```

Output:

